

The Context

Antimicrobial Resistance (AMR) is a global health challenge, that in 2019 was directly responsible for **1.27 million deaths** [1]. Cytecom [2], a Warwick-based start-up, is working to tackle AMR by developing a rapid and reliable device capable of detecting viable bacteria in under a minute. Their approach combines fluorescent microscopy with electrical stimulation to achieve quick and accurate results.

The Problem

We want to provide a statistical framework for classifying the membrane-potential dynamics that is **rigorous** and **scalable**. This can be used as a proxy for the **proliferation capacity** of cells, as described in the pioneering research in [3].

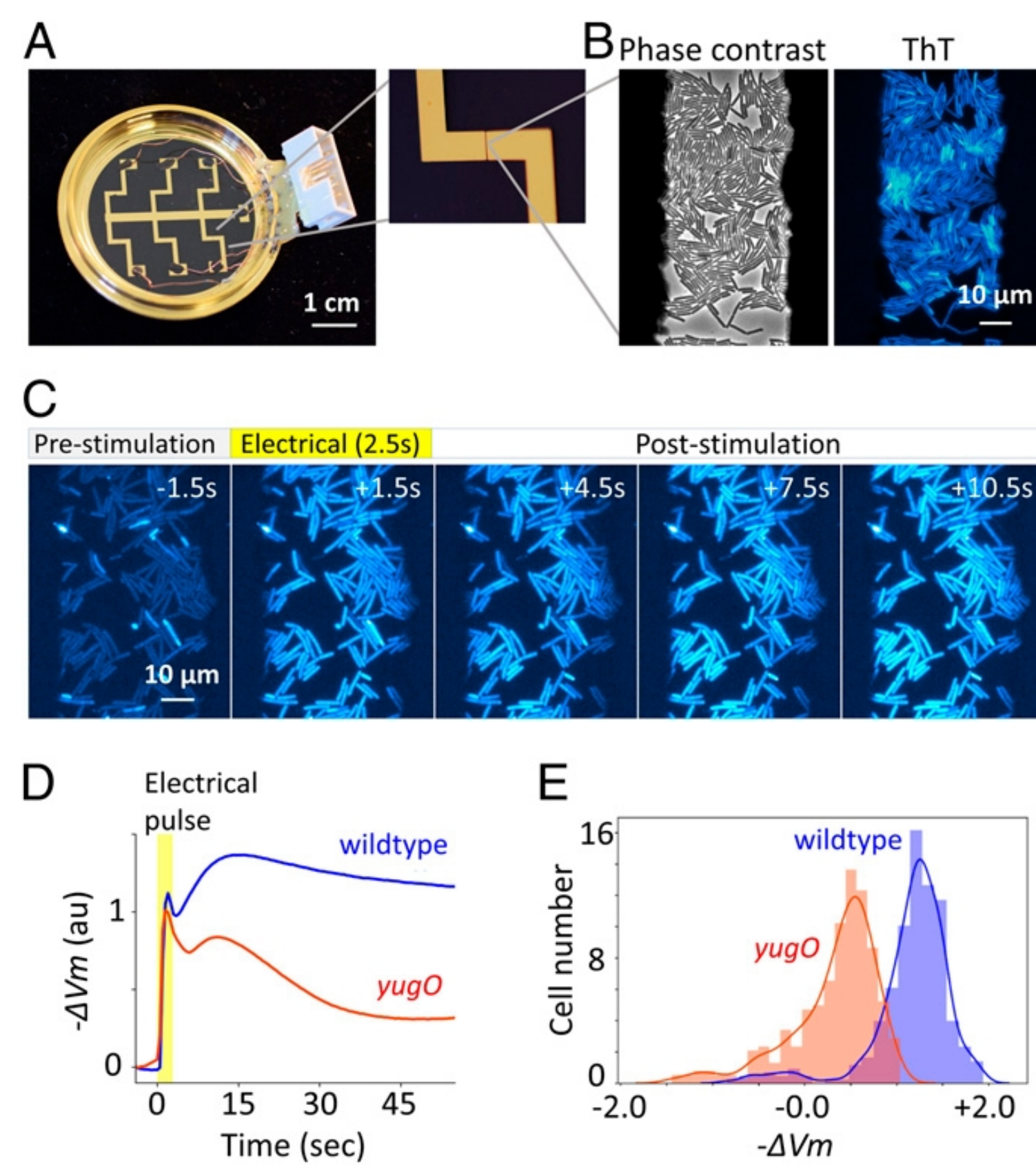


Figure 1. From [3], (A) Custom electrode-coated dish with 50-μm gap delivers electrical pulses. (B) Cells imaged via phase-contrast and thioflavin T (ThT) fluorescence. (C) ThT signal increases during stimulation, indicating hyperpolarization. (D) Time course of $-\Delta V_m$ for Wildtype and *yugO* strains. (E) Histogram of $-\Delta V_m$ at 30 s post-stimulation shows distinct responses.

Our Modelling Approach

In [3] a **FitzHugh-Nagumo (FHN)** process is used to model the **membrane potential** (V_t) and the recovery variable (W_t) of bacterial cells, after undergoing electrical stimulation. We extend this model to a **stochastic FHN** to account for:

- Model error
- Observation error
- Stochasticity in the system

The sFHN is as follows:

$$\frac{dV_t}{dt} = k \left((V_t + V_0) - \alpha (V_t + V_0)^3 + W_t \right) + \frac{dI_t^v}{dt} + \sigma_m^v \frac{dB_t^v}{dt} \quad (1)$$

$$\frac{dW_t}{dt} = -((V_t + V_0) + 1 - 0.1 \log k - W_t) + \frac{dI_t^w}{dt} + \sigma_m^w \frac{dB_t^w}{dt}, \quad (2)$$

where:

- k can be used to differentiate proliferative and inhibited cells
- α controls the dynamic range of membrane potential
- I represents the electrical stimulus
- We assume we observe $\tilde{V}_t = V_t + \sigma_o \cdot \mathcal{N}(0, 1)$

We thus are required to classify k for paths like those in Figure 2:

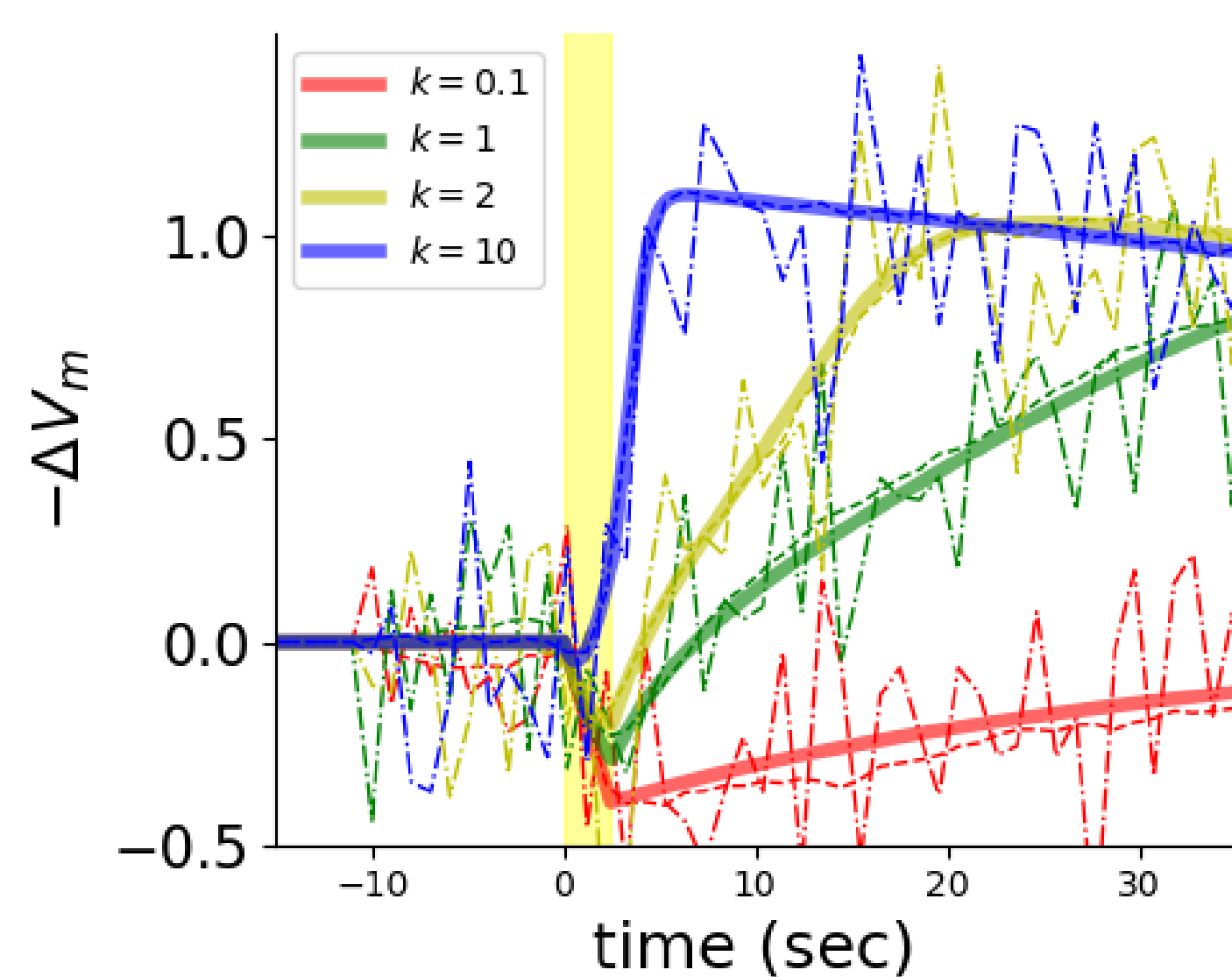


Figure 2. Solid lines indicate the solutions to the deterministic FHN model. Dashed lines indicate additions of the 2 sources of noise.

Due to the resultant application, we have the following requirements for any statistical approach to classification:

- Accuracy
- Robustness
- Physical Interpretability
- Speed (at the point of use)

Simulated Dataset

Ongoing technological development required us to create a replica dataset to test the signature regression on, as the model parametrisation may change over time. This allowed us to tailor our method to be robust to mixed data types. We create a spatial dataset where we can independently vary the clustering of cells, and then the clustering of **proliferation** within the simulation.

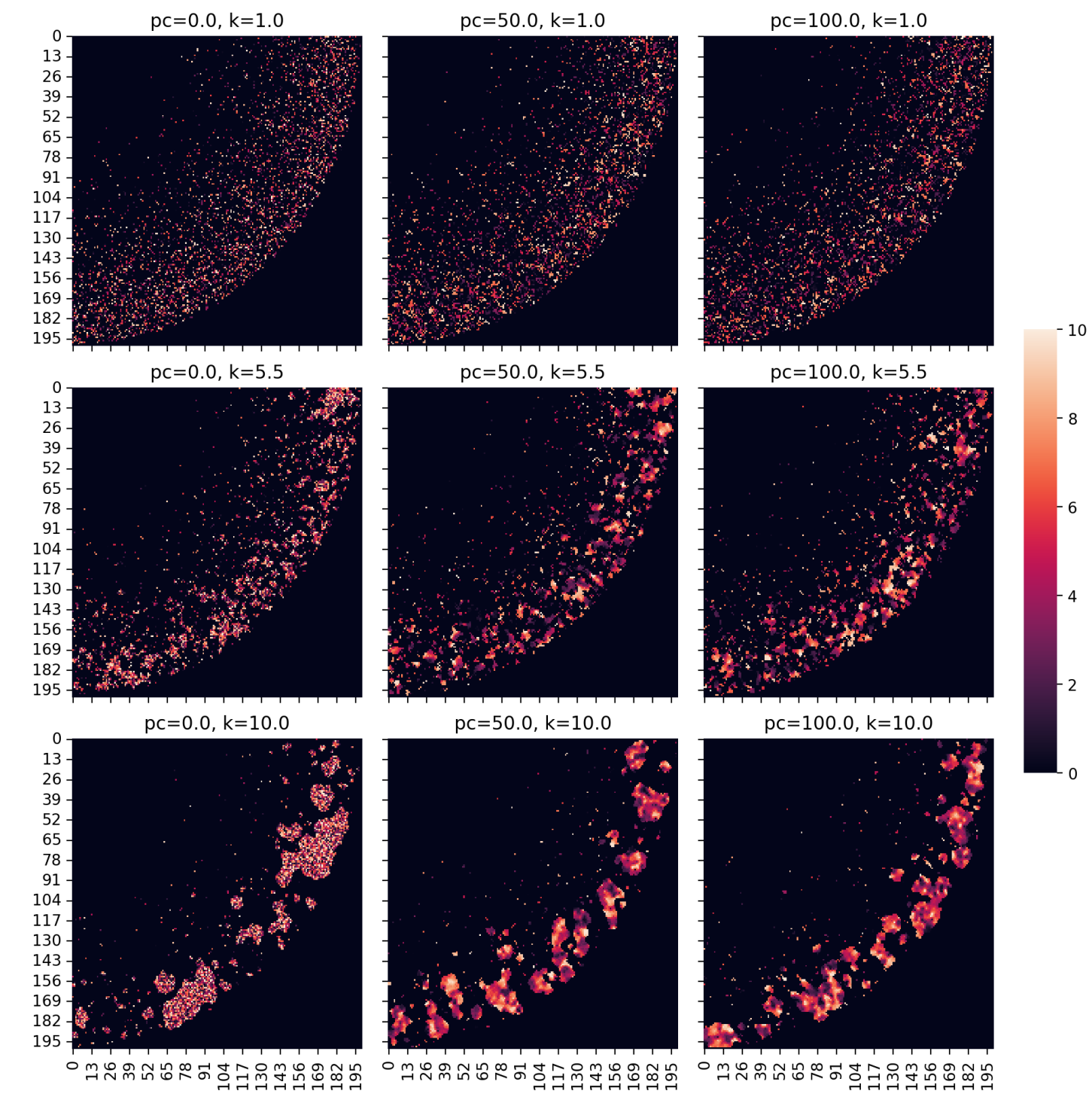


Figure 3. An example simulated data set for different values of Preferential Constant (pc) and a 'clumping' parameter k .

The Path Signature - A Geometric Feature Space

The signature of a path X_t , $S(X)$, is a **geometric** feature set computed by taking iterated integrals of X_t (see [4] for more details):

$$S(X)_{[0,T]} = (1, \mathbf{X}^1, \mathbf{X}^2, \dots), \text{ where } \mathbf{X}^n = \int_{0 < u_1 < u_2 < \dots < u_n < T} \bigotimes_{i=1}^n dX_{u_i}.$$

The signature **linearises functions of the path** X_t [5], rendering it an appropriate feature set for **regression**.

The simulated dataset allowed us to gather results on robustness to other distributions.

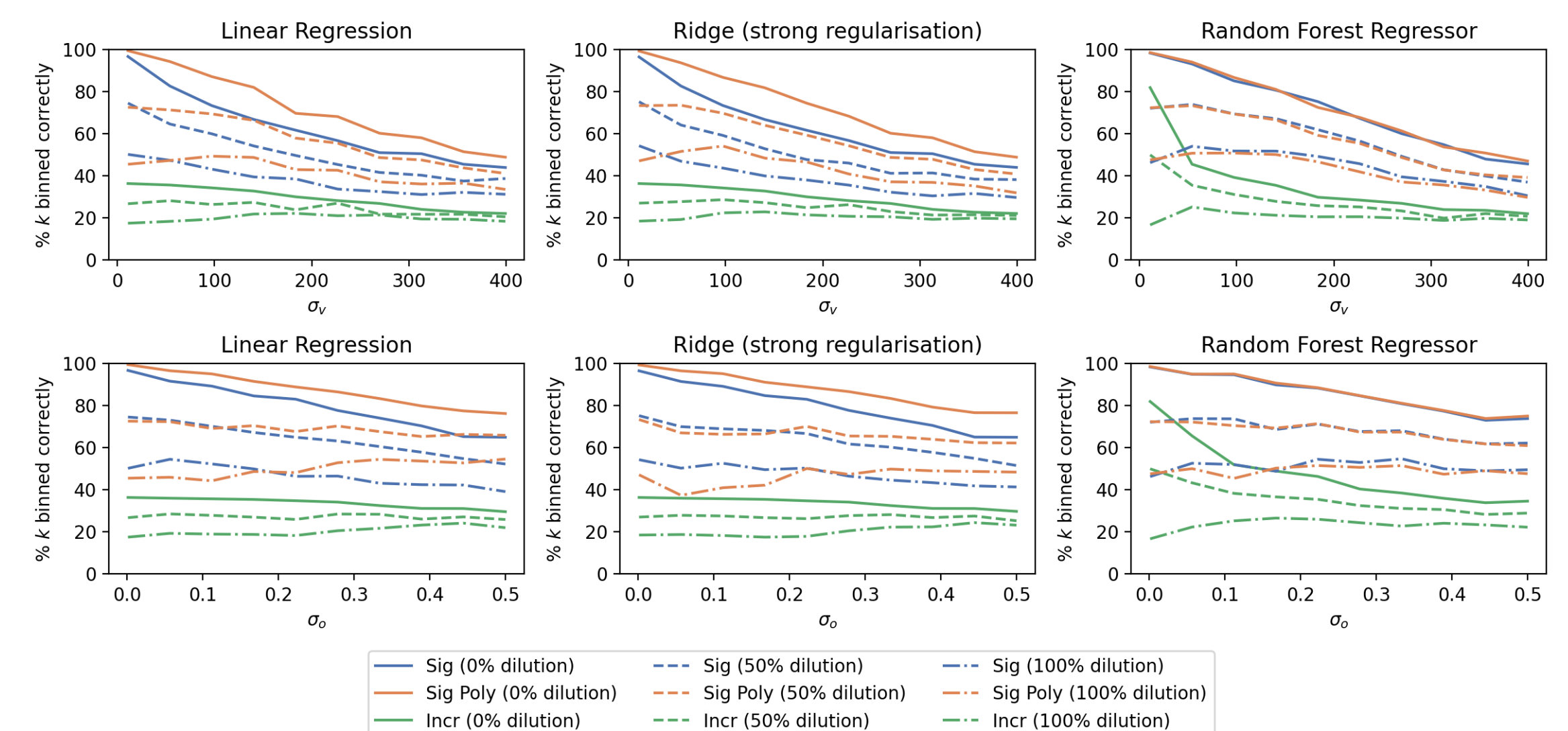


Figure 4. Classification accuracy across different dilutions of the test data, across different specifications of noise

Ongoing Work - Inference using Signatures

Moving forward, we would like to generalise these results, and give theoretical guarantees. We assume we observe the output to some 'expensive' differential equation (3), and have an auxiliary simulator model that is 'cheap' (4):

$$Y_t = f_\theta(Y_t) \cdot dX_t \quad (3)$$

$$\tilde{Y}_t = \tilde{f}_\theta(\tilde{Y}_t) \cdot d\tilde{X}_t \quad (4)$$

If we use linear regression on signatures to estimate the parameters θ using the simulations, i.e. we estimate

$$\hat{\theta}(S(\tilde{Y})) = \beta S(\tilde{Y}) + \epsilon,$$

we would like to make the distance $d(\hat{\theta}(S(\tilde{Y})), \theta)$ concrete and compare it to other methods, such as MLE or simulation based approaches, such as ABC. We keep the assumptions on f and X relatively general (i.e. non-Brownian).

Acknowledgements and References

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- (1) Antimicrobial Resistance Collaborators, *Lancet*, 2022, **399**, 629–655.
- (2) Cytecom, <https://www.cytecom.co.uk/>, Accessed: May 2025.
- (3) J. P. Stratford, C. L. A. Edwards, M. J. Ghanshyam, D. Malyshev, M. A. Delise, Y. Hayashi and M. Asally, *Proceedings of the National Academy of Sciences*, 2019, **116**, 9552–9557.
- (4) T. Lyons, *Rough paths, Signatures and the modelling of functions on streams*, 2014.
- (5) A. Fermanian, *Journal of Multivariate Analysis*, 2022, **192**, 105031.